

PROFESSIONAL SUMMARY

Computational Scientist and Machine Learning Engineer (M.Sc. GPA: 3.98/4.0) specializing in building highly optimized, scalable ML pipelines and custom clustering algorithms. Proven expertise in utilizing Python, Cython, and Linux HPC to extract reliable insights from massive, high-dimensional datasets. Adept at bridging rigorous mathematical formulations with robust software engineering to solve complex data challenges.

EDUCATION

•Master of Science in Computational Sciences

Institut Teknologi Bandung, Indonesia GPA: 3.98/4.0
– Start Date: 9 September 2024
– Graduation Date: 12 August 2026

•Bachelor of Science in Astronomy

Institut Teknologi Bandung, Indonesia GPA: 3.25/4.0
– Start Date: 17 September 2020
– Graduation Date: 24 August 2024

EXPERIENCE

•Research Intern – ASPIRE Program

University of Amsterdam Amsterdam, The Netherlands
June 2026 – August 2026
– **HPC Numerical Simulation:** Built a scalable Python simulation pipeline from first principles on HPC infrastructure to model complex dynamical systems and test alternative hypotheses.
– **Time-Series Optimization:** Implemented robust parameter estimation on synthetic time-series data using Box-Least Squares periodograms, multi-start optimization, and differential evolution algorithms (GitHub).

Tech Stack: Python, Time-Series Analysis, HPC, Numerical Simulation, Optimization.

•Machine Learning Researcher (Graduate) – P2MI

Institut Teknologi Bandung Bandung, Indonesia
September 2024 – August 2026
– **Algorithm Optimization (GUMM):** Developed a custom Expectation-Maximization (EM) mixture model optimized in **Cython** with **OpenMP parallelization** to significantly accelerate spatial data processing; featured in conference proceedings paper and open-sourced on GitHub.
– **Statistical Pipeline Architecture:** Architected a robust pipeline integrating hierarchical density-based clustering, Xtreme Deconvolution GMM, and Kernel Density Estimation to process large-scale, high-dimensional datasets.
– **Hybrid ML Pipeline (ENTRAP):** Engineered a custom unsupervised machine learning pipeline combining density-based clustering with Topological Data Analysis (TDA), improving recall by 13–94% across 10 million-scale datasets while maintaining precision at 1.00; packaged with a standard Scikit-Learn style API (GitHub).

Tech Stack: Python, Cython, OpenMP, Bayesian Statistics, Unsupervised ML, Topological Data Analysis, Scikit-Learn API.

•Machine Learning Teaching Assistant

Institut Teknologi Bandung Bandung, Indonesia
February 2025 – June 2026
– **Technical Mentorship:** Instructed graduate students on the end-to-end Machine Learning lifecycle, encompassing data preprocessing, model selection, and supervised vs. unsupervised paradigms.
– **Curriculum Design:** Designed hands-on engineering assignments covering Deep Learning (CNNs) and classic ML algorithms utilizing **PyTorch**, **TensorFlow**, and **Scikit-Learn**.

Tech Stack: Python, TensorFlow, PyTorch, Scikit-Learn, Deep Learning.

•Data Pipeline Engineer – High-Energy Physics Project

Institut Teknologi Bandung Bandung, Indonesia
September 2024 – November 2025

- **Data Pipeline Execution:** Deployed a data-reduction and signal-processing pipeline on a **Linux HPC cluster** to effectively clean and extract features from highly noisy, low-signal data streams.
- **Statistical Modeling:** Conducted advanced statistical analysis and mathematical fitting to extract key physical parameters, with results published in a reputable **Q1 peer-reviewed journal** (MNRAS).

Tech Stack: Linux HPC, Signal Processing, Data Reduction, Statistical Modeling, Python.

SKILLS

- **Programming Languages:** Python, SQL, LaTeX, C/C++, Cython.
- **Machine Learning & AI:** TensorFlow, PyTorch, Keras, Scikit-Learn, Deep Learning, Bayesian Statistics, Unsupervised/Supervised Learning.
- **Data Engineering & HPC:** Linux HPC, OpenMP, MPI (mpi4py, mpich), Pandas, NumPy, SciPy, Large-Scale Data Processing, Data Pipelines.
- **Tools & Systems:** Git, Linux/Windows, AMUSE, Scikit-Learn API Design.

Languages

- **Indonesian:** Native
- **English:** High Intermediate

COURSES AND CERTIFICATION

•Online Machine Learning Course

Certification from freeCodeCamp – Focus on Machine Learning and Neural Networks

August 2024

- TensorFlow and Keras for Neural Networks.
- Gradient Descent Techniques and Data Preparation.
- Classification with Neural Networks.
- Natural Language Processing with Scikit-Learn and TensorFlow.

•Online Data Science Course

Certification from DQ Lab – Focus on Data Exploration and Visualization

July - August 2021

- Exploratory Data Analysis with Python for Beginners.
- Data Visualization with Python (Matplotlib) for Beginners.
- COVID-19 Data Analysis in Indonesia using Python.
- Python for Data Professionals (Beginners).
- Data Science Training: Data Exploration and Visualization using Python.

ACHIEVEMENTS

•Dean's Award FMIPA ITB, Semester I - 2023/2024

February 2024

Awarded for academic excellence at the Faculty of Mathematics and Natural Sciences.

GPA: 4.00